EXPERIMENT 8

Determination and plotting of Capacitance of Co-axial cable per unit length with different (i) inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

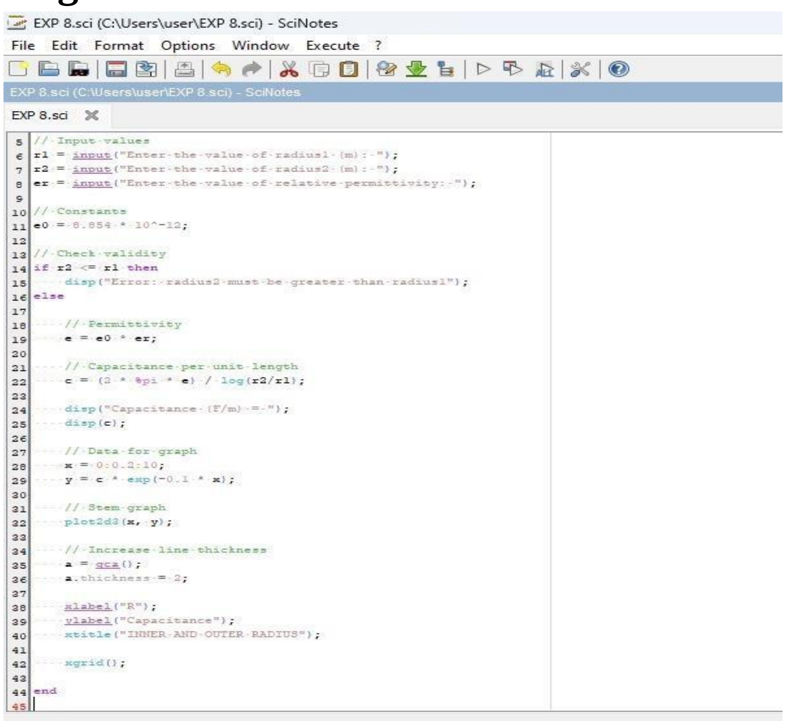
Aim:

To write program for Capacitance of co-axial cable per unit length using SCILAB.

Software required:

•SCILAB version 6.0.1

Program:


EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes

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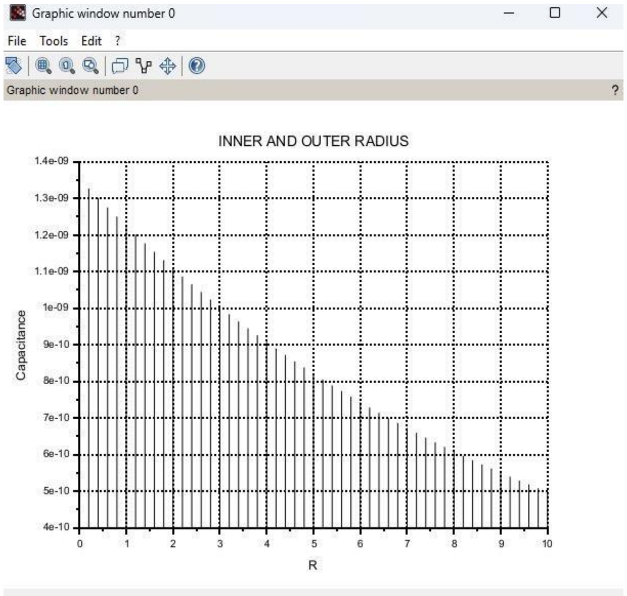
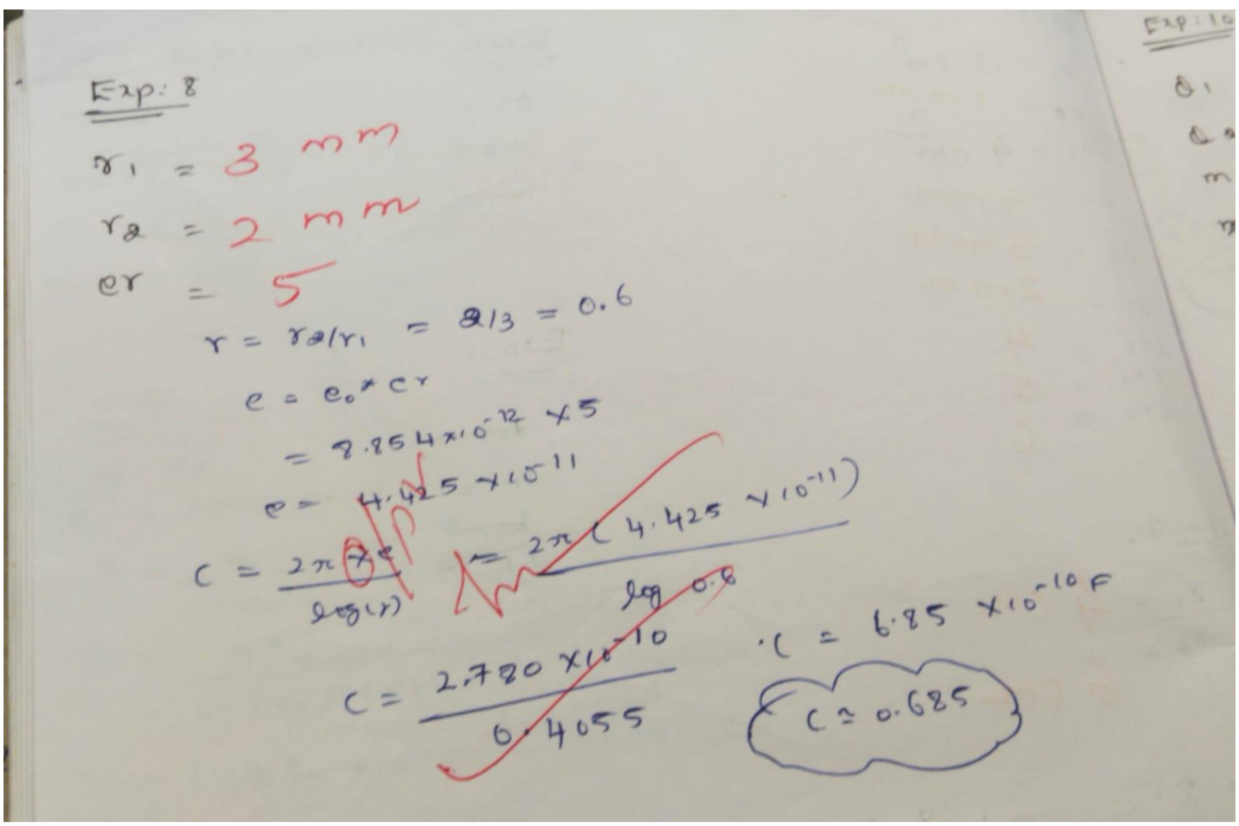
EXP 8.sci

```
// Input values
r1 = input("Enter the value of radius1 (m): ");
r2 = input("Enter the value of radius2 (m): ");
er = input("Enter the value of relative permittivity: ");

// Constants
e0 = 8.854 * 10^-12;

// Check validity
if r2 <= r1 then
    disp("Error: radius2 must be greater than radius1");
else

    // Permittivity
    e = e0 * er;

    // Capacitance per unit length
    c = (2 * %pi * e) / log(r2/r1);

    disp("Capacitance (F/m) = ");
    disp(c);

    // Data for graph
    x = 0:0.2:10;
    y = c * exp(-0.1 * x);

    // Stem graph
    plot2d3(x, y);

    // Increase line thickness
    a = gca();
    a.thickness = 2;

    xlabel("R");
    ylabel("Capacitance");
    xtitle("INNER AND OUTER RADIUS");

    xgrid();

end
```

Output:



Practical value:

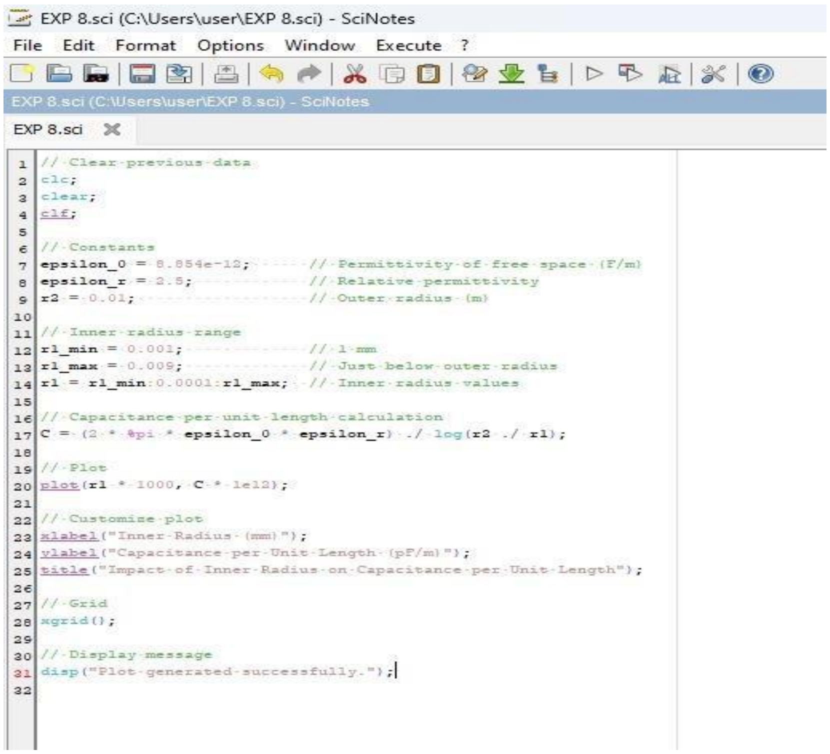


```
Enter the value of radius1 (m): 3

Enter the value of radius2 (m): 4

Enter the value of relative permittivity: 7

 "Capacitance (F/m) = "
  1.354D-09
```

CASE STUDY 1:

Program:

Variation with Inner Radius

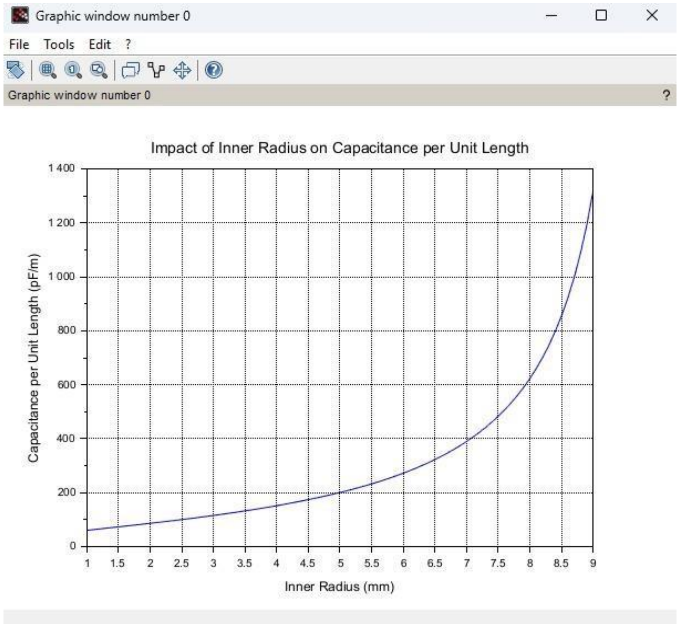



```
// Clear previous data
clc;
clear;
clf;

// Constants
epsilon_0 = 8.854e-12;      // Permittivity of free space (F/m)
epsilon_r = 2.5;            // Relative permittivity
r2 = 0.01;                  // Outer radius (m)

// Inner radius range
r1_min = 0.001;             // 1 mm
r1_max = 0.009;             // Just below outer radius
r1 = r1_min:0.0001:r1_max;  // Inner radius values

// Capacitance per unit length calculation
C = (2 * %pi * epsilon_0 * epsilon_r) ./ log(r2 ./ r1);

// Plot
plot(r1 * 1000, C * 1e12);

// Customize plot
xlabel("Inner Radius (mm)");
ylabel("Capacitance per Unit Length (pF/m)");
title("Impact of Inner Radius on Capacitance per Unit Length");

// Grid
xgrid();

// Display message
disp("Plot generated successfully.");
```

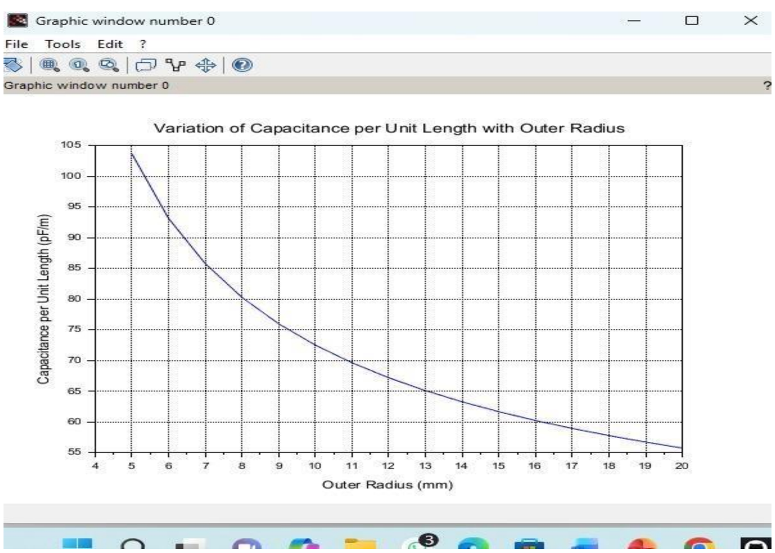
Output:

CASE STUDY 2:

Program:



Variation with Outer Radius



CASE STUDY 3:

Program:

EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes

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EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes

EXP 8.sci

```
// Clear the console and workspace
clc;
clear;
clf;

// Constants
epsilon0 = 8.854e-12;    // Permittivity of free space (F/m)
epsilon_r = 3.0;         // Relative permittivity
a = 1e-3;                // Inner radius (1 mm)

// Outer radius values (5 mm to 20 mm)
b_values = 5e-3:1e-3:20e-3;

// Initialize capacitance array
C_prime = zeros(b_values);

// Calculate capacitance per unit length
for i = 1:length(b_values)
    b = b_values(i);
    epsilon = epsilon0 * epsilon_r;
    C_prime(i) = (2 * %pi * epsilon) / log(b / a);
end

// Plot the results
scf(0);
plot(b_values * 1e3, C_prime * 1e12);

// Labels and title
xlabel("Outer Radius (mm)");
ylabel("Capacitance per Unit Length (pF/m)");
title("Variation of Capacitance per Unit Length with Outer Radius");

// Display grid
xgrid();

// Display message
disp("Plot generated successfully.");
```

Output:

Variation with Relative Permittivity

CASE STUDY 4:

Program:

EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes

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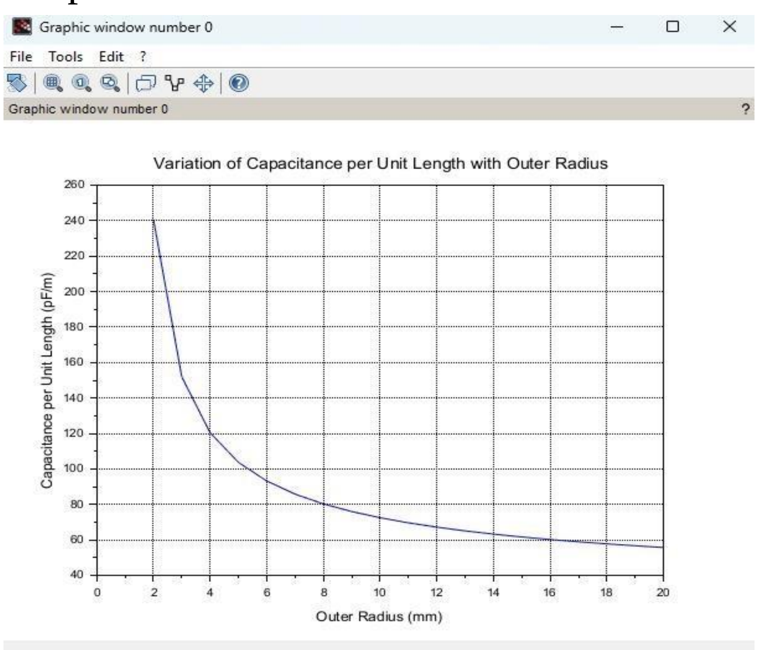
EXP 8.sci (C:\Users\user\EXP 8.sci) - SciNotes

EXP 8.sci

```
clc;
clear;
clf;

// Constants
epsilon0 = 8.854e-12;
epsilon_r = 3;
a = 1e-3;                  // Inner radius (1 mm)

// Outer radius from 1 mm to 20 mm
b = 1e-3:1e-3:20e-3;

C = zeros(b);

for i = 1:length(b)
    if b(i) > a then
        C(i) = (2 * %pi * epsilon0 * epsilon_r) / log(b(i)/a);
    else
        C(i) = %nan;      // Ignore b=a
    end
end

// Plot
scf(0);
plot(b*1000, C*1e12);

xlabel("Outer Radius (mm)");
ylabel("Capacitance per Unit Length (pF/m)");
title("Variation of Capacitance per Unit Length with Outer Radius");

xgrid();
```

Output:



Result :

Thus the program was verified for Capacitance of co-axial cable per unit length using SCILAB was successfully.